

Healthcare Predictive Analytics

Project Title: Drug Demand Forecasting Using Historical Pharmacy Sales Data

Objective: To develop a basic predictive analytics project using historical healthcare data and a simple statistical model to forecast future medicine demand. The project uses simulated pharmacy sales data for educational purposes.

1. Problem Selection

The selected problem is **drug demand forecasting**. Pharmacies need to estimate future medicine requirements so that commonly used medicines remain available while excess inventory and wastage are reduced. Historical monthly sales can be used as a simple predictor of near-term demand.

2. Dataset Description

A simulated 12-month dataset was created for three commonly stocked medicines. The variables are month, medicine, units sold and month number. The dataset is intended to demonstrate the workflow of predictive analytics and does not represent real patient or pharmacy records.

Month	Paracetamol	Amoxicillin	Metformin
Jan	420	180	260
Feb	435	190	270
Mar	450	205	278
Apr	465	215	285
May	480	225	295
Jun	495	235	305
Jul	510	245	315
Aug	530	255	325
Sep	545	265	335
Oct	560	280	345
Nov	580	295	358
Dec	600	310	370

3. Data Exploration

- **Highest annual sales:** Paracetamol has the highest total simulated sales (5,570 units).
- **Second:** Metformin has 3,841 units across the year.
- **Third:** Amoxicillin has 2,600 units across the year.
- **Trend:** All three medicines show an increasing monthly sales pattern in the simulated dataset.
- **Potential use:** The increasing pattern can be used to estimate the next month's requirement.

4. Predictive Model

A simple **linear regression** model is used. Month number (1–12) is the independent variable and monthly units sold are the dependent variable. The model estimates a straight line through the historical observations and can then be used to forecast the next month.

Medicine	Regression equation	Forecast for Month 13	R ²
Paracetamol	$y = 400.8 + 16.2x$	611 units	0.998
Amoxicillin	$y = 167.8 + 11.4x$	316 units	0.995
Metformin	$y = 247.6 + 9.9x$	376 units	0.996

5. Model Evaluation

The coefficient of determination (R^2) is used as a basic measure of how closely the linear model follows the historical data. Values closer to 1 indicate that the linear trend explains a larger proportion of the variation in the simulated observations. Because this is a small educational dataset with an intentionally increasing trend, the model is expected to show a strong fit. In real pharmacy forecasting, additional variables and out-of-sample validation would be required.

6. Forecast Interpretation

The model forecasts approximately **622 units of Paracetamol**, **324 units of Amoxicillin**, and **380 units of Metformin** for month 13. These values are mathematical estimates based only on the simulated historical trend; they should not be treated as actual purchasing recommendations.

7. Tools and Workflow

- Dataset creation and organization using spreadsheet-style tabular data.
- Data cleaning and checking for missing or inconsistent values.
- Exploratory analysis of monthly medicine sales.
- Linear regression for trend-based forecasting.
- Evaluation using R^2 and comparison of predicted versus historical values.
- Presentation of findings in a structured report.

8. Limitations

- The dataset is simulated and relatively small.
- The model uses only time/month as a predictor.
- Seasonality, promotions, disease outbreaks, stock-outs and prescribing patterns are not included.
- Real-world forecasting should use separate training and testing data and appropriate validation.
- Forecasts should support, not replace, professional inventory and clinical decision-making.

9. Conclusion

This project demonstrates a basic healthcare predictive analytics workflow: selecting a problem, preparing historical data, exploring trends, building a simple predictive model, evaluating the model and interpreting a forecast. Linear regression provides an easy-to-understand starting point for drug demand forecasting, while more advanced models can be considered when larger and more detailed real-world datasets are available.

Note: This report uses simulated data solely for academic/project demonstration.